

Titanic Survival Analysis: A Closer Examination

An In-Depth Look at the Factors Affecting Survival Rates During the Titanic Disaster



Tools Used

✓ My SQL

✓ CANVA



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Titanic Survival Analysis

Summary

The analysis explores the Titanic dataset to identify patterns in survival outcomes based on demographic, economic, and social factors. It covers data exploration, survival analysis, fare and travel patterns, family/group dynamics, and advanced insights like passenger categories and group survival trends.

Key Points

Data Exploration

- Examined first 10 passengers with details (Name, Age, Sex, Pclass, Survived).
- Counted total passengers.
- Computed average age by passenger class.

Survival Analysis

- Overall survival vs non-survival distribution.
- Survival rate by gender (females had higher survival).
- Survival rate by passenger class (1st class had better survival chances).
- Survival count by embarkation port.

Economics & Travel

- Analyzed fare: maximum, minimum, and average fare per class.
- Total fare collected from each embarkation port.
- Identified class that paid the highest average fare.

Family & Group

- Created new column $\text{FamilySize} = \text{SibSp} + \text{Parch} + 1$.
- Identified passengers traveling alone.

Advanced Insights

- Categorized passengers: Child, Adult, Senior.
- Compared survival counts across these categories.
- Determined gender + class combination with highest survival rate.
- Identified passengers sharing same ticket number (likely families/friends).

Conclusion

- Gender and Class were the strongest predictors of survival.
- Women and children, especially in 1st class, had the best chances of survival.
- Men in 3rd class had the lowest chances of survival.
- Fare and Embarked Port acted as indirect indicators of wealth and class, which heavily influenced outcomes.
- The disaster highlights how social inequalities and demographic factors determined survival in a life-and-death situation.

Titanic Survival Analysis

Data Exploration

First 10 passengers with their Name, Age, Sex, Pclass, Survived

```
1 -- First 10 passengers with their Name, Age, Sex, Pclass, Survived
2 • SELECT Name, Age, Sex AS Gender, Pclass AS Passengers_Class, Survived from sql_project.titanic LIMIT 10 ;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
Name	Age	Gender	Passengers_Class	Survived
Braund, Mr. Owen Harris	22	male	3	0
Cumings, Mrs. John Bradley (Florence Briggs Th...	38	female	1	1
Heikkinen, Miss. Laina	26	female	3	1
Futrelle, Mrs. Jacques Heath (Lily May Peel)	35	female	1	1
Allen, Mr. William Henry	35	male	3	0
McCarthy, Mr. Timothy J	54	male	1	0
Palsson, Master. Gosta Leonard	2	male	3	0
Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	27	female	3	1
Nasser, Mrs. Nicholas (Adele Achem)	14	female	2	1
Sandstrom, Miss. Marguerite Rut	4	female	3	1

Total passengers in the dataset.

```
1 -- Count total passengers in the dataset.
2 • SELECT count(*) AS total_number_of_passengers FROM sql_project.titanic;
```

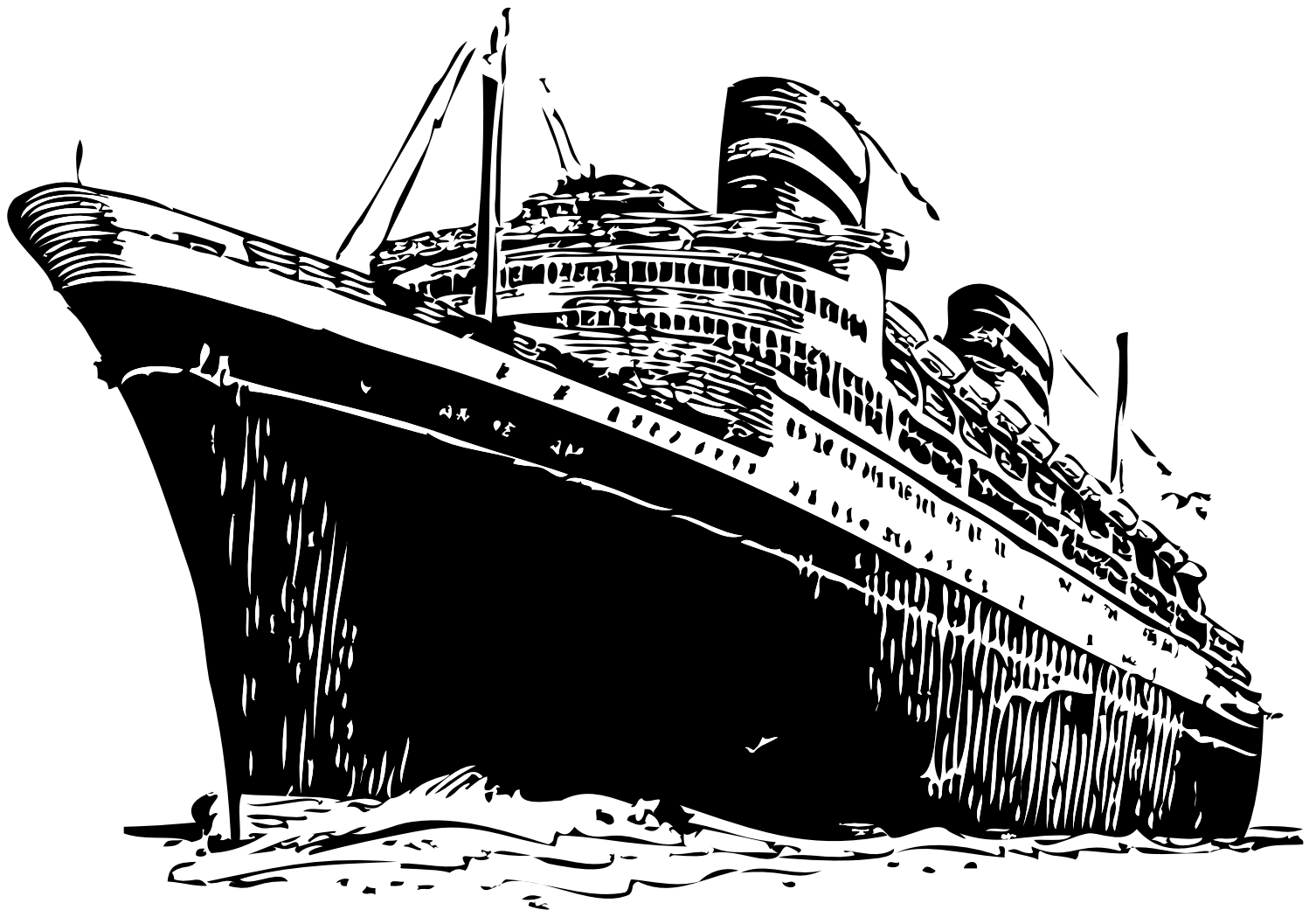
Result Grid	Filter Rows:	Export:	Wrap Cell Content:
total_number_of_passengers			
714			

Average age of passengers by Passenger_class

```
1 -- Calculate average age of passengers by Pclass
2 SELECT Pclass AS passengers_class, ROUND(AVG(Age),2) AVG_Age FROM sql_project.titanic GROUP BY Pclass ORDER BY Pclass;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [FA](#)

	passengers_class	AVG_Age
▶	1	38.24
	2	29.89
	3	25.16



Survival Analysis

Survived vs did not survive

Limit to 300 rows

```
1  -- Count how many survived vs did not survive
2  • SELECT Survived, COUNT(Survived) AS total_count FROM sql_project.titanic GROUP BY Survived
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Survived	total_count
▶	0	424
	1	290

Survival rate by gender

Limit to 300 rows

```
1  -- Calculate survival rate by gender
2  • SELECT Sex AS Gender, ROUND(AVG(Survived)*100,2) AS Survival_Rate_Percentage
3  FROM sql_project.titanic GROUP BY Sex;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Gender	Survival_Rate_Percentage
▶	male	20.53
	female	75.48

Survival rate by Passenger_class.

The screenshot shows a SQL IDE with multiple tabs at the top: "SQL File 11*", "titanic", "SQL File 9*", "titanic", "SQL File 5*", and "SQL File 6* x". The active tab is "SQL File 6*", which contains the following SQL query:

```
1  -- Calculate survival rate by Passenger_class
2  • SELECT Pclass AS Passenger_class, ROUND(AVG(Survived)*100,2) AS Survival_Rate_Percentage
3  FROM sql_project.titanic GROUP BY Pclass ORDER BY Passenger_class;
```

Below the query editor, the "Result Grid" is displayed, showing the output of the query. It has columns "Passenger_class" and "Survival_Rate_Percentage". The data is as follows:

Passenger_class	Survival_Rate_Percentage
1	65.59
2	47.98
3	23.94

Survival count by Embarked port.

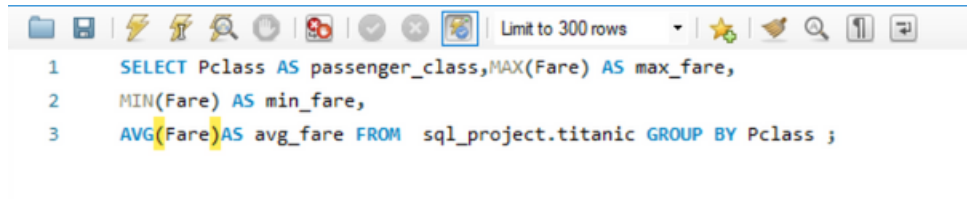
```
1 -- Find survival count by Embarked port
2 SELECT Embarked As Embarked_port ,COUNT(Survived) AS Total_Count FROM sql_project.titanic WHERE Survived = "1" GROUP BY Embarked;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

Embarked_port	Total_Count
C	79
S	201
O	8

Economics & Travel Analysis

The maximum, minimum, and average Fare by passenger class.

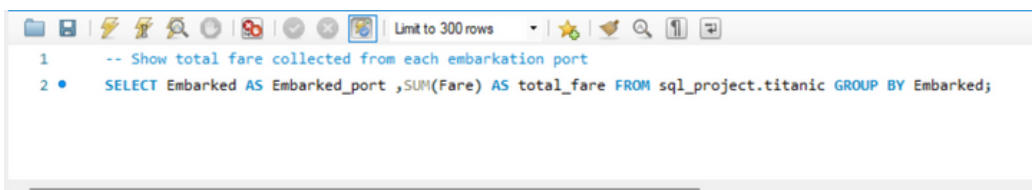


```
1 SELECT Pclass AS passenger_class, MAX(Fare) AS max_fare,
2 MIN(Fare) AS min_fare,
3 AVG(Fare) AS avg_fare FROM sql_project.titanic GROUP BY Pclass ;
```

Result Grid

	passenger_class	max_fare	min_fare	avg_fare
1	1	512.3292	0	87.96158225806447
2	2	73.5	10.5	21.47155606936416
3	3	56.4958	0	13.229435211267623

Show total fare collected from each embarkation port.

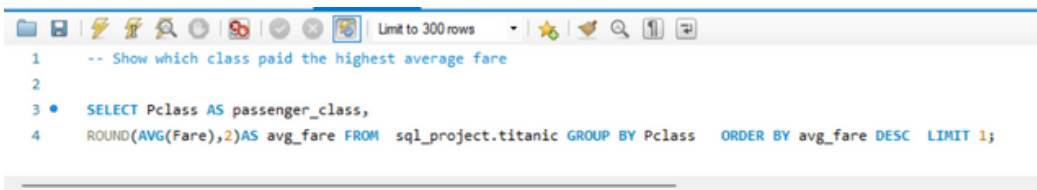


```
1 -- Show total fare collected from each embarkation port
2 • SELECT Embarked AS Embarked_port , SUM(Fare) AS total_fare FROM sql_project.titanic GROUP BY Embarked;
```

Result Grid

	Embarked_port	total_fare
1	S	15221.86159999984
2	C	8878.5797
3	Q	511.4417
4	160	

Class paid the highest average fare.



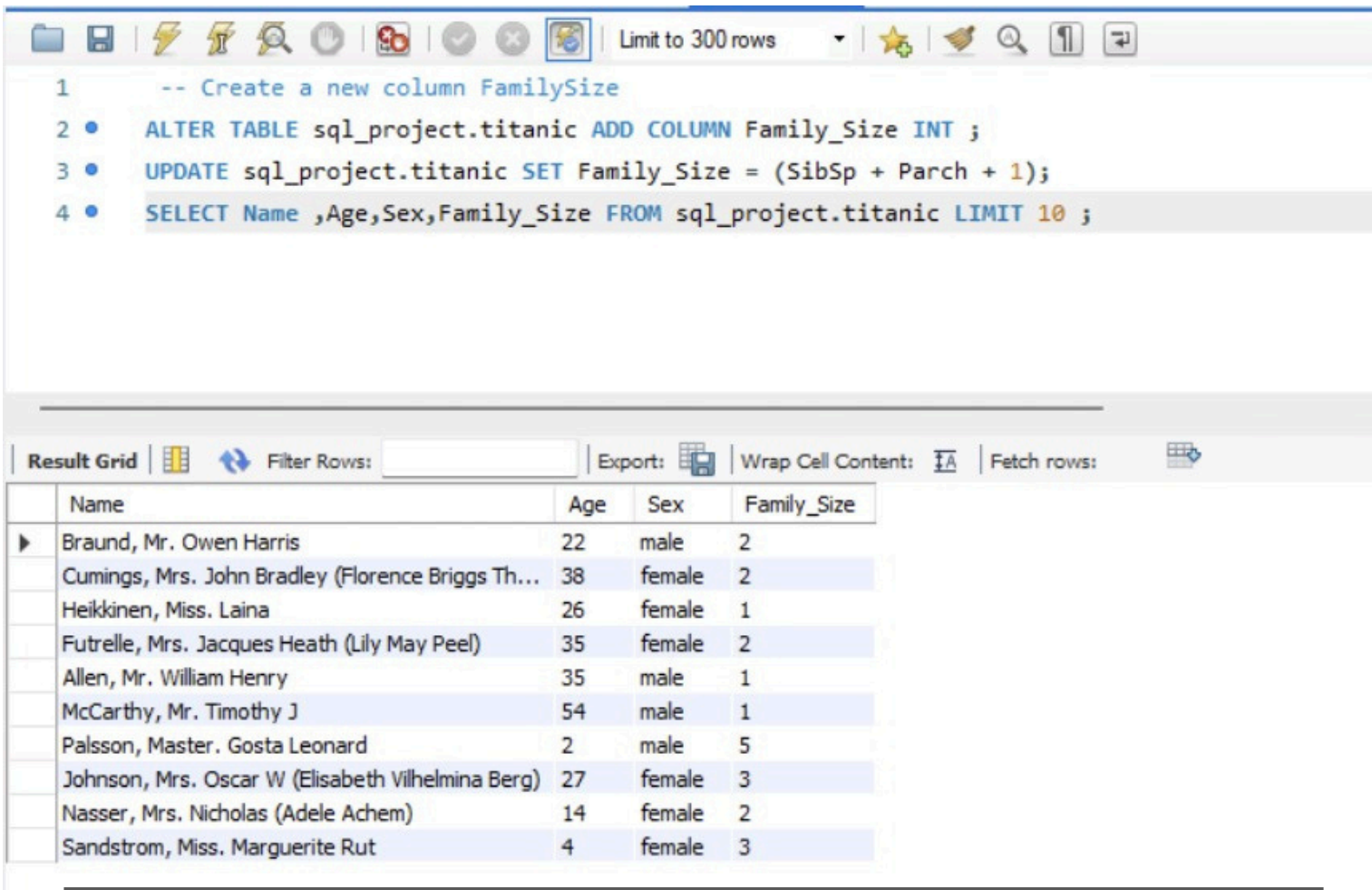
```
1 -- Show which class paid the highest average fare
2
3 • SELECT Pclass AS passenger_class,
4 ROUND(AVG(Fare),2) AS avg_fare FROM sql_project.titanic GROUP BY Pclass ORDER BY avg_fare DESC LIMIT 1;
```

Result Grid

	passenger_class	avg_fare
1	1	87.96

Family & Group Analysis

A new column FamilySize = SibSp + Parch



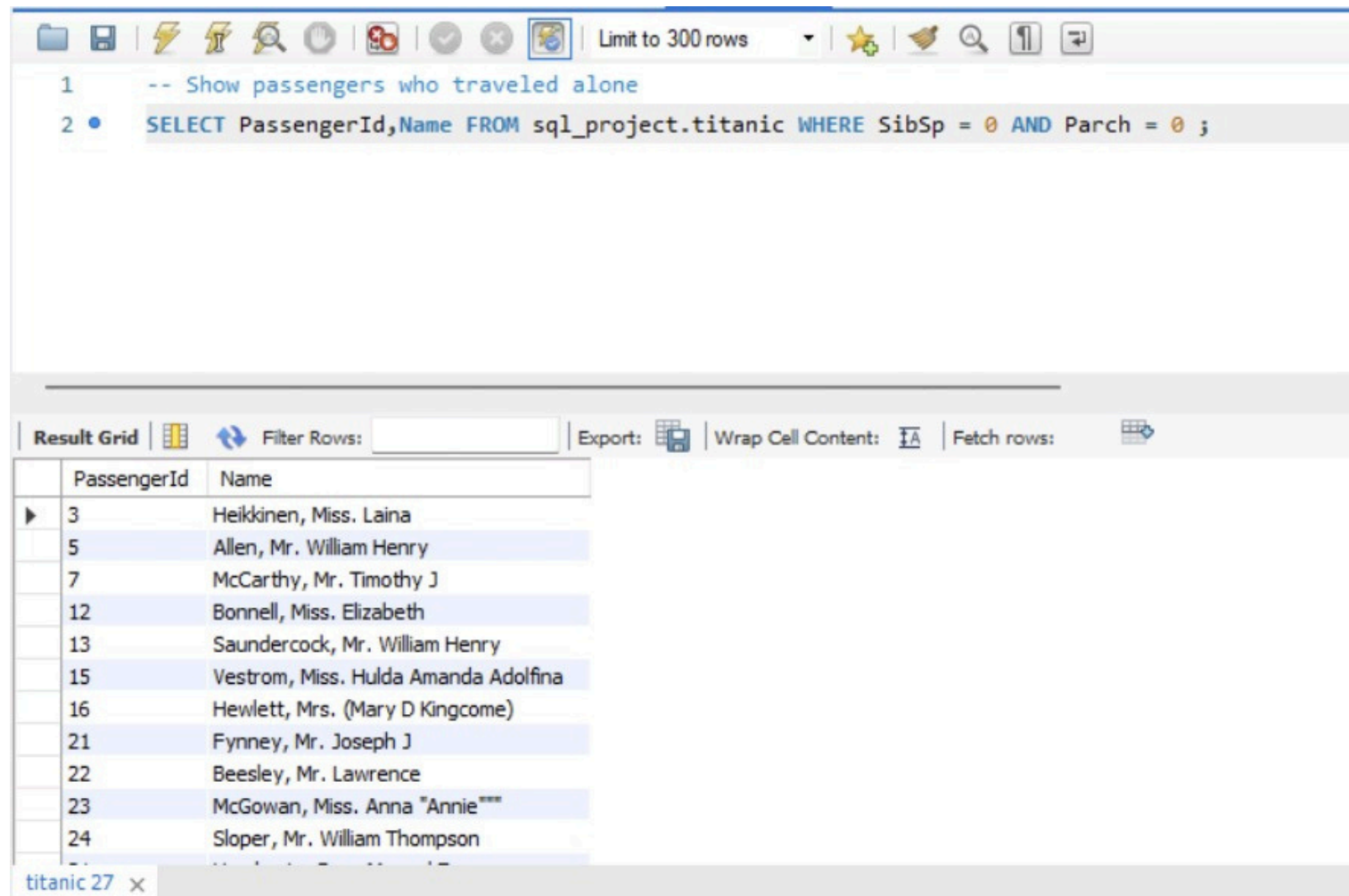
The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, a search icon, and a 'Limit to 300 rows' dropdown. The SQL editor contains the following code:

```
1 -- Create a new column FamilySize
2 • ALTER TABLE sql_project.titanic ADD COLUMN Family_Size INT ;
3 • UPDATE sql_project.titanic SET Family_Size = (SibSp + Parch + 1);
4 • SELECT Name ,Age,Sex,Family_Size FROM sql_project.titanic LIMIT 10 ;
```

Below the editor is the 'Result Grid' tab. It shows a table with 5 columns: Name, Age, Sex, and Family_Size. The table contains 10 rows of data.

Name	Age	Sex	Family_Size
Braund, Mr. Owen Harris	22	male	2
Cumings, Mrs. John Bradley (Florence Briggs Th...	38	female	2
Heikkinen, Miss. Laina	26	female	1
Futrelle, Mrs. Jacques Heath (Lily May Peel)	35	female	2
Allen, Mr. William Henry	35	male	1
McCarthy, Mr. Timothy J	54	male	1
Palsson, Master. Gosta Leonard	2	male	5
Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	27	female	3
Nasser, Mrs. Nicholas (Adele Achem)	14	female	2
Sandstrom, Miss. Marguerite Rut	4	female	3

Passengers who traveled alone



The screenshot shows a SQL IDE interface. The top toolbar is identical to the previous one. The SQL editor contains the following code:

```
1 -- Show passengers who traveled alone
2 • SELECT PassengerId,Name FROM sql_project.titanic WHERE SibSp = 0 AND Parch = 0 ;
```

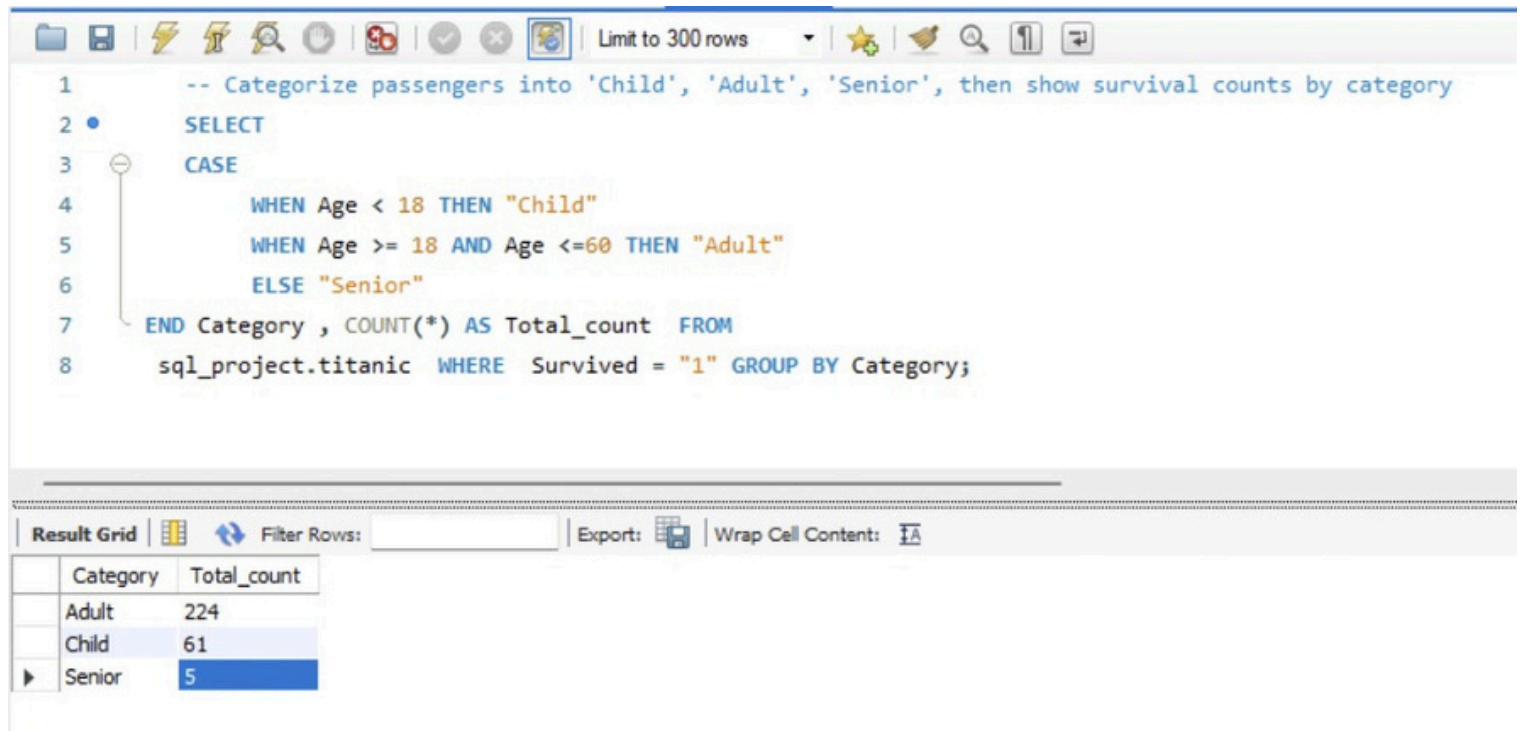
Below the editor is the 'Result Grid' tab. It shows a table with 2 columns: PassengerId and Name. The table contains 12 rows of data.

PassengerId	Name
3	Heikkinen, Miss. Laina
5	Allen, Mr. William Henry
7	McCarthy, Mr. Timothy J
12	Bonnell, Miss. Elizabeth
13	Saunders, Mr. William Henry
15	Vestrom, Miss. Hulda Amanda Adolfina
16	Hewlett, Mrs. (Mary D Kingcome)
21	Fynney, Mr. Joseph J
22	Beesley, Mr. Lawrence
23	McGowan, Miss. Anna "Annie" "
24	Sloper, Mr. William Thompson

At the bottom left, there is a tab labeled 'titanic 27' with a close button (X).

Advanced Insights

Categorize passengers into 'Child', 'Adult', 'Senior', then show survival counts by category.

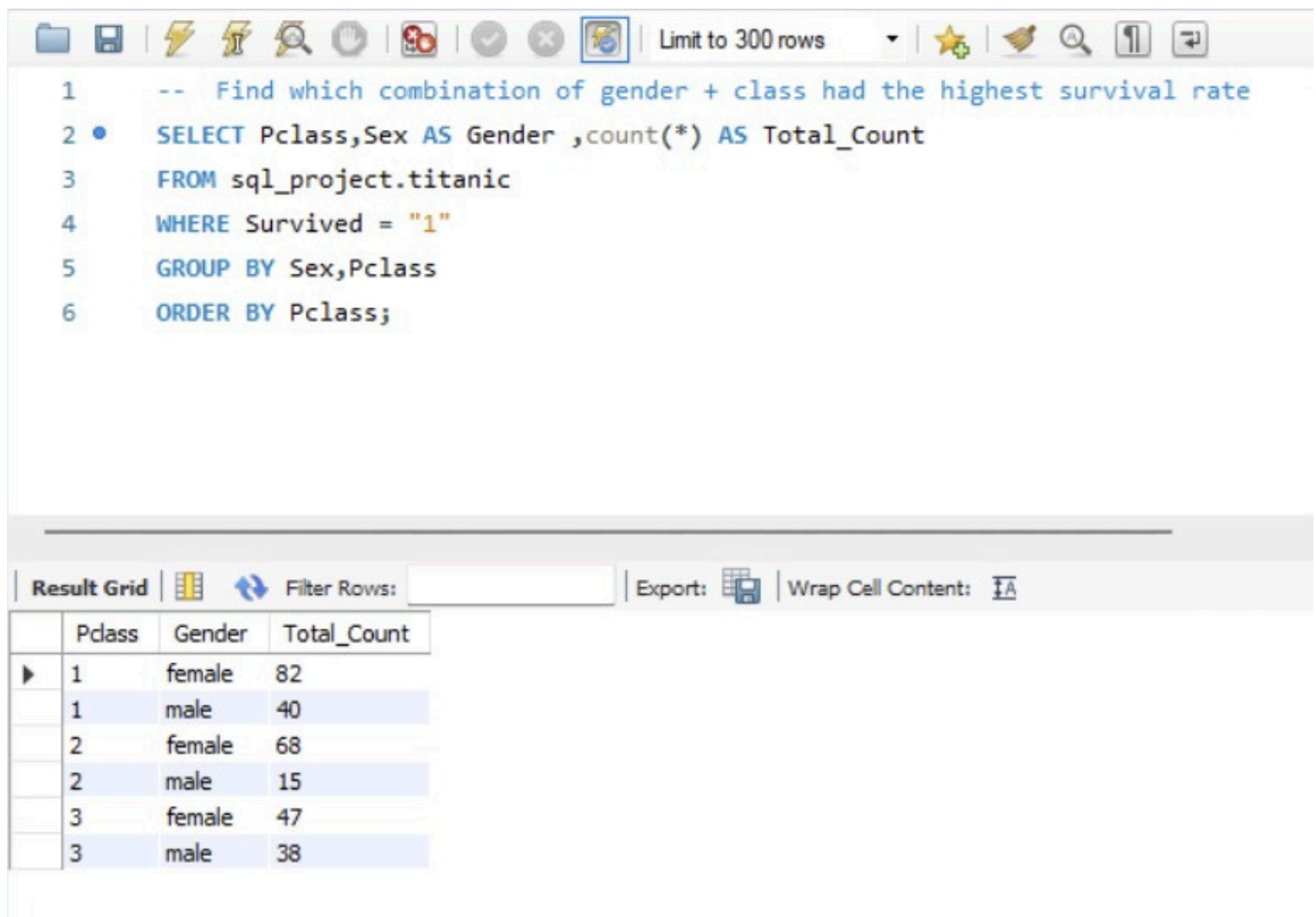


```
1 -- Categorize passengers into 'Child', 'Adult', 'Senior', then show survival counts by category
2 SELECT
3 CASE
4     WHEN Age < 18 THEN "Child"
5     WHEN Age >= 18 AND Age <=60 THEN "Adult"
6     ELSE "Senior"
7 END Category , COUNT(*) AS Total_count FROM
8 sql_project.titanic WHERE Survived = "1" GROUP BY Category;
```

Result Grid

Category	Total_count
Adult	224
Child	61
Senior	5

Combination of gender + class had the highest survival rate

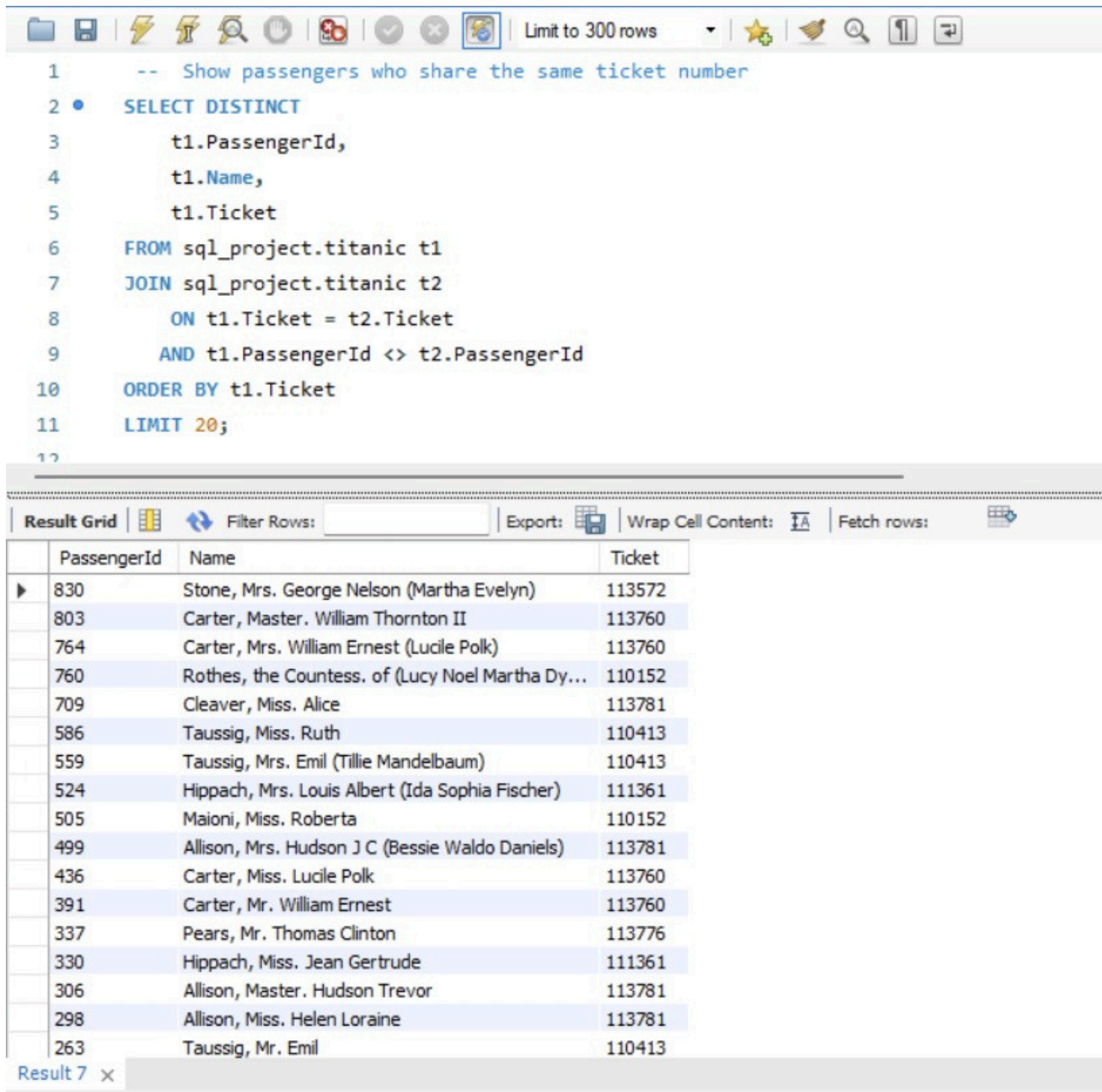


```
1 -- Find which combination of gender + class had the highest survival rate
2 SELECT Pclass,Sex AS Gender ,count(*) AS Total_Count
3 FROM sql_project.titanic
4 WHERE Survived = "1"
5 GROUP BY Sex,Pclass
6 ORDER BY Pclass;
```

Result Grid

Pclass	Gender	Total_Count
1	female	82
1	male	40
2	female	68
2	male	15
3	female	47
3	male	38

Passengers who share the same ticket number (possible family/friends groups)



The screenshot shows a SQL IDE interface. At the top, there is a toolbar with various icons and a dropdown menu set to "Limit to 300 rows". Below the toolbar, a SQL query is entered in a text area. The query is as follows:

```
-- Show passengers who share the same ticket number
SELECT DISTINCT
    t1.PassengerId,
    t1.Name,
    t1.Ticket
FROM sql_project.titanic t1
JOIN sql_project.titanic t2
    ON t1.Ticket = t2.Ticket
    AND t1.PassengerId <> t2.PassengerId
ORDER BY t1.Ticket
LIMIT 20;
```

Below the query editor, there is a "Result Grid" section. It includes a "Filter Rows:" input field, an "Export:" button, a "Wrap Cell Content:" checkbox, and a "Fetch rows:" button. The results are displayed in a table with the following columns: PassengerId, Name, and Ticket.

PassengerId	Name	Ticket
830	Stone, Mrs. George Nelson (Martha Evelyn)	113572
803	Carter, Master. William Thornton II	113760
764	Carter, Mrs. William Ernest (Lucile Polk)	113760
760	Roths, the Countess. of (Lucy Noel Martha Dy...	110152
709	Cleaver, Miss. Alice	113781
586	Taussig, Miss. Ruth	110413
559	Taussig, Mrs. Emil (Tillie Mandelbaum)	110413
524	Hippach, Mrs. Louis Albert (Ida Sophia Fischer)	111361
505	Maioni, Miss. Roberta	110152
499	Allison, Mrs. Hudson J C (Bessie Waldo Daniels)	113781
436	Carter, Miss. Lucile Polk	113760
391	Carter, Mr. William Ernest	113760
337	Pears, Mr. Thomas Clinton	113776
330	Hippach, Miss. Jean Gertrude	111361
306	Allison, Master. Hudson Trevor	113781
298	Allison, Miss. Helen Loraine	113781
263	Taussig, Mr. Emil	110413

At the bottom left of the results section, it says "Result 7 x".

Thank You
Kumar Udeshya